You can solve this equation in a number of ways. For example, you can solve it numerically on a computer. Start with a known value of $\psi(x)$ and update it slightly by calculating the derivative. Once you have the derivative, calculate how $\psi(x)$ changes in a small increment of time. Then, add this incremental change to $\psi(x)$ and keep doing it over and over. It turns out that $\psi(x)$ will do some interesting things—it will move around somehow. In fact, under certain circumstances, it will form a wave packet that moves around very much like a harmonic oscillator.

10.4 ENERGY LEVELS

The other thing you can do with the Hamiltonian is calculate the energy levels of the oscillator, by finding the energy eigenvectors and eigenvalues. As we learned in Lecture 4, once you know these eigenvectors and eigenvalues, you can figure out the time dependence without solving any differential equations. That's because you already know the time dependence of each energy eigenvector. You may want to review the Schrödinger's Ket recipe we gave in Section 4.13.

For now, let's concentrate on finding the energy eigenvectors themselves, using the time-independent Schrödinger equation:

$$\mathbf{H}|\psi_E\rangle = E|\psi_E\rangle.$$

The subscript E indicates that ψ_E is the eigenvector for a particular eigenvalue E. This equation defines two things: the wave functions $\psi_E(x)$ and the energy levels E. Let's make things less abstract by expanding **H** using Eq. 10.12:

$$-\frac{\hbar^2}{2}\frac{\partial^2\psi_E(x)}{\partial x^2} + \frac{1}{2}\omega^2 x^2 \psi_E(x) = E\psi_E(x). \qquad (10.14)$$

To solve this equation, we must: